https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	(sin) θ is the independent variable and v is the dependent variable **or** vary (sin) θ and measure v	**1**
	keep s (PQ) constant	**1**
	Methods of data collection	
	labelled diagram showing inclined plane with labelled support **and** P and Q marked	**1**
	method to measure angle e.g. use a protractor to measure θ or use a ruler to measure marked distances from which sin θ or θ may be determined	**1**
	method of timing for an appropriate distance to determine v (at Q) e.g. use a stopwatch/timer **or** correctly positioned light gate(s) connected to a timer/data-logger **or** correctly positioned motion sensor connected to data-logger	**1**
	measurement of an appropriate distance to determine v (at Q) e.g. rule to measure an appropriate length **or** length of a card to interrupt light beam **or** distance from motion sensor to Q	**1**
	Method of analysis	
	plot a graph of v^2 against sin θ	**1**
	relationship valid if a straight line produced (not passing through the origin)	**1**
	$g = -\text{gradient} \times \frac{B+m}{2ms}$ **or** $g = y\text{-intercept} \times \frac{B+m}{2Bs}$	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max. 6**
	D1 use cushion/foam/sandbox for falling body (*B*)	
	D2 (sin) θ determined using trigonometry relationship using marked lengths	
	D3 appropriate equation to determine *v* (at Q) e.g. $v = \frac{2s}{t}$	
	D4 repeat experiment for each θ and average *v* or *t*	
	D5 use of balance to measure mass of wooden block *m* and falling body *B* and rule to measure *s*	
	D6 *y*-intercept $= \frac{2Bsg}{B+m}$.	
	D7 clean surfaces of blocks/inclined plane/ensure surface of the plane is smooth	
	D8 keep *B* and *m* constant or keep mass of block and mass of falling body constant	
	D9 method to ensure that wooden block starts at the same position P, e.g. put a mark on the block or align front or back of block	
	D10 method to prevent plane slipping so that angle being measured remains the same, e.g. a mass as a stop	

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